

Alex Bergman

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🌐 <http://alex-lukas-bergman.github.io/>



Occupation

2026 – ... 📌 **Postdoctoral researcher** Massachusetts institute for technology. I am a Knut and Alice Wallenberg postdoctoral researcher in harmonic analysis, complex analysis, and partial differential equations. My host is Professor Aleksandr Logunov.
You can find my page on the Knut and Alice Wallenberg stiftelse homepage here, [KAW](#).

2021 – 2026 📌 **PhD student** Lund university. I defended my thesis on the 5th of June 2026. My advisor was Professor Alexandru Aleman.

Research Publications

Preprints

- 1 A. Aleman and A. Bergman, *Invariant subspaces for generalized differentiation and Volterra operators*, 2025. arXiv: [2503.08443 \[math.FA\]](#). 🔗 URL: <https://arxiv.org/abs/2503.08443>.
- 2 A. Bergman, *A remark on the Beurling-Malliavin theorem in several variables*, 2025. arXiv: [2512.07271 \[math.CV\]](#). 🔗 URL: <https://arxiv.org/abs/2512.07271>.

Journal Articles

- 1 A. Bergman, "Inner approximations of doubling weights with applications to Beurling-Malliavin theory in Toeplitz kernels," *Trans. Amer. Math. Soc.*, 2026. 🔗 DOI: <https://doi.org/10.1090/tran/9844>.
- 2 A. Bergman, "A sharp entropy condition for the density of angular derivatives," *C. R. Math. Acad. Sci. Paris*, vol. 363, pp. 29–33, 2025, ISSN: 1631-073X, 1778-3569. 🔗 DOI: [10.5802/crmath.695](https://doi.org/10.5802/crmath.695).
- 3 A. Bergman, "Hörmander's inequality and point evaluations in de Branges space," *Rev. Mat. Iberoam.*, 2025. 🔗 DOI: <https://doi.org/10.4171/rmi/1582>.
- 4 A. Bergman, "On cyclicity in de Branges-Rovnyak spaces," *Indiana Univ. Math. J.*, vol. 73, no. 4, pp. 1307–1329, 2024, ISSN: 0022-2518, 1943-5258. 🔗 DOI: [10.1512/iumj.2024.73.60011](https://doi.org/10.1512/iumj.2024.73.60011).
- 5 A. Bergman and O. Rubin, "Chebyshev polynomials corresponding to a vanishing weight," *J. Approx. Theory*, vol. 301, Paper No. 106048, 25, 2024, ISSN: 0021-9045, 1096-0430. 🔗 DOI: [10.1016/j.jat.2024.106048](https://doi.org/10.1016/j.jat.2024.106048).

Academic Talks

2026 📌 **Which functions belong to a given model space?**, Conference "Math bench sessions", Krakow, Poland.

📌 **Interpolation in Toeplitz kernels**, Joint Mathematics Meeting, Washington D.C.

2025 📌 **Hörmander's inequality in de Branges space**, Analysis seminar, Bilkent university

Academic Talks (continued)

- **Hörmander's inequality in de Branges space**, Analysis seminar, Lund university
- **Hörmander's inequality in de Branges space**, Conference "Hilbert Function Spaces", Rome, Italy.
- **Hörmander's inequality in de Branges space**, Conference "Math bench sessions", Krakow, Poland.
- 2024 ■ **Hörmander's inequality in de Branges space**, Analysis seminar, KU Leuven, Belgium.
- **A sharp entropy condition for the density of angular derivatives**, Conference "Operators and analytic function spaces", Luminy, France.
- **Smooth invariant chains in de Branges space**, Conference "Advanced courses in operator theory and complex analysis", Tenerife, Spain.
- **Invariant subspaces for generalized differentiation and Volterra operators**, Analysis seminar, NTNU, Norway.
- **Invariant subspaces for generalized differentiation and Volterra operators**, Analysis seminar, King's college London, England.
- **Cyclic Vectors in de Branges-Rovnyak Spaces**, Analysis seminar, Lund university, Sweden.
- 2023 ■ **Smooth invariant chains in de Branges space**, Conference "Math bench sessions", Krakow, Poland.
- **Invariant subspaces for second order ODE**, Analysis seminar, Stockholm university, Sweden.
- **Invariant subspaces for second order ODE**, Conference "International Conference on Spectral Theory and Approximation", Lund, Sweden.
- **On cyclicity in de Branges-Rovnyak spaces**, Conference "Advanced courses in operator theory and complex analysis", Thessaloniki, Greece.
- 2022 ■ **Interpolation spaces between H^1 and H^∞** , Math seminar, Umeå university, Sweden.

Grants

- 2026 ■ **Knut and Alice Wallenberg postdoctoral researcher grant** Postdoctoral research grant. [KAW](#).
Knut och Alice Wallenbergs stiftelse.
- 2024 ■ **The Fund of the Torsten and Fanny Brodén Foundation** General research grant.
The Royal Physiographic Society of Lund.
- **The Bokelund Society Travel Grant** Travel Grant.
Faculty of Science, Lund university.

Grants (continued)

- 2023  **The Fund of the Walter Gyllenberg Foundation** General research grant.
The Royal Physiographic Society of Lund.
- 2022-2024  **Thorild Dahlgrens och Folke Lannérs stipendiefond** Travel grant.
Awarded several times.

Academic Visits

Research Stays

- 2024  **Research stay hosted by Profesor Alexei Poltoratski**, November 2024, University of Wisconsin, Wisconsin, USA.
-  **Research stay hosted by Professor Alexander Pushnitski**, May 2024, King's college, London, England

Summer Schools

- 2023  **Analysis, arithmetics and geometry: zeta-functions, trace formulas, and around**, Mittag-Leffler institute, Stockholm, Sweden.
- 2022  **Summer school in Harmonic analysis**, Bonn/Kopp, Germany.

Teaching

- 2026  **Algebra and vector geometry** Spring, Lund university.
-  **Linear algebra** Spring, Lund university.
- 2025  **Linear algebra** Fall, Lund university.
-  **Linear analysis** Fall, Lund university.
-  **Envariabelanalys (analysis in one variable)** Spring, Lund university.
- 2024  **Analysis in several variables I** Fall, Lund university.
-  **Analysis in one variable** Fall, Lund university.
-  **Envariabelanalys (analysis in one variable)** Spring, Lund university.
- 2023  **Analysis in one variable** Fall, Lund university.
-  **Envariabelanalys (analysis in one variable)** Spring, Lund university.
- 2022  **Analysis in several variables II** Fall, Lund university.

Teaching (continued)

■ **Analysis in one variable** Fall, Lund university.

■ **Envariabelanalys (analysis in one variable)** Spring, Lund university.

2021 ■ **Analysis in one variable** Fall, Lund university.

In addition, I have been a substitute lecturer in several courses at the Bachelor and Master level including:

Analysis I

Analysis in several variables I

Analysis in several variables II

Linear analysis

Functional analysis

Partial differential equations

Referee Work

I have been a referee for the following journals.

Journal of Functional Analysis

Analysis and Mathematical Physics

Education

2018-2021 From 2018 to 2021 I was a student at the department of mathematics at Lund university.

Master thesis (2021): *Interpolation spaces between Hilbert and Banach spaces of analytic functions*

Bachelor thesis (2020): *Weak type spectral mapping theorems for strongly continuous groups of operators*

Community outreach and Popular Science

2024 **Forskaruppsats** I participated in an interview study for high school students at "Polhemskolan" (Lund, Sweden) aspiring to be scientists.

Question and answer session. I answered questions from high school students in Skåne, Sweden about mathematics over the phone.

2023 **Question and answer session.** I answered questions from high school students in Skåne, Sweden about mathematics over the phone.

Other departmental duties

I was a member of the following hiring committees

2025 PhD student in dynamical systems at Lund university.

2023 PhD student in representation theory at Lund university.

I am or have been a member of the following boards.

2024-... I am currently a member of the Mathematics program council at Lund university. The purpose of the council is education development. My role is as the PhD student representative.